CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

# Chapter 12: Surface Areas and Volumes

Previous year board questions, topic-wise: 15 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Surface area

1. Find the total surface area of a solid hemisphere whose diameter is $2d$. (a) $3\pi d^2$ (b) $2\pi d^2$ (c) $\frac{1}{2}\pi d^2$ (d) $\frac{3}{4}\pi d^2$ CBSE 2026, 1 mark

2. Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius $r$ is $3\pi r^2$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2025, 1 mark

3. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) $1:1$ (b) $1:4$ (c) $2:3$ (d) $3:2$ CBSE 2024, 1 mark

4. A cylinder 5 cm high has a curved surface area of 94.2 cm². Find its radius, taking $\pi = 3.14$. (a) 2 cm (b) 3 cm (c) 2.9 cm (d) 6 cm CBSE 2023, 1 mark

5. Find the curved surface area of a cone of height 24 cm and radius 7 cm. (a) 528 cm² (b) 1056 cm² (c) 550 cm² (d) 500 cm² CBSE 2023, 1 mark

6. A conical cavity of height 24 cm is hollowed out of a solid cylinder of height 30 cm and radius 7 cm. The cone has the same radius as the cylinder. Find the total surface area of the solid that is left. CBSE 2022 Term 2, 4 marks

7. A wooden article is made by scooping a hemisphere out of each end of a solid cylinder. The cylinder is 10 cm high and its base has radius 3.5 cm. Find the total surface area of the article. Take $\pi = \frac{22}{7}$. CBSE 2018, 3 marks

8. A tent is a cylinder with a cone of the same diameter on top. The cylinder is 2.1 m high and 3 m in diameter, and the slant height of the cone is 2.8 m. Canvas costs ₹500 per square metre. Find the cost of the canvas for the tent. Take $\pi = \frac{22}{7}$. CBSE 2016, 3 marks

9. After a flood, 50 schools agree to share equally the cost of canvas for 1500 tents. Each tent is a cylinder of base radius 2.8 m and height 3.5 m, with a cone of the same radius and height 2.1 m on top. Canvas costs ₹120 per square metre. Find the amount each school pays. Take $\pi = \frac{22}{7}$. CBSE 2016, 4 marks

## 2 Volume

**10.** A solid is a cylinder with a hemisphere on each end. The solid is 20 cm long overall and the cylinder is 7 cm in diameter. Find the volume of the solid. Take $\pi = \frac{22}{7}$. CBSE 2026, 3 marks

**11.** A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21}$ m³ of air. Find the height of the cylindrical part. Take $\pi = \frac{22}{7}$. CBSE 2025, 3 marks

**12.** The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm CBSE 2024, 1 mark

**13.** A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm³. Find the two radii. CBSE 2024, 3 marks

**14.** A room is a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21}$ m³ of air. Find the total height of the room. Take $\pi = \frac{22}{7}$. CBSE 2023, 3 marks

**15.** An ice-cream cone has radius 3 cm and height 12 cm. The lowest part of the cone, one-sixth of its volume, is left empty. The rest is filled with ice-cream, with a hemisphere of ice-cream on top. Find the volume of the ice-cream. Take $\pi = 3.14$. CBSE 2023, 3 marks

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